Advanced Flying Robots – Final Results

INDI Across Three Platforms: Standard, Thrust-Upgraded, Brushless CF21BL

Georg Wolnik

Final Presentation

Agenda

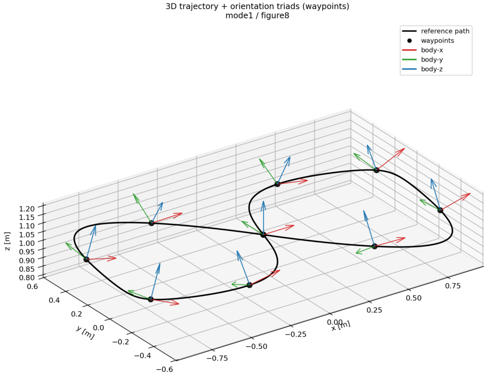
1. Recap: planned trajectories + geometric vs. INDI (standard drone)
2. Core result: figure-8 across all four modes
3. Upgraded vs. brushless: oval, circle, slalom + kt sweeps
4. Limitations & complications
5. Summary

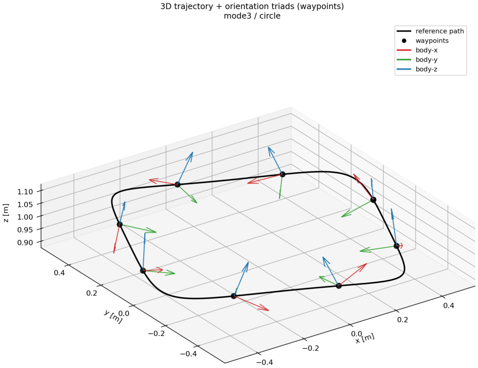
Recap — Motion Planning Pipeline

From the last progress presentation

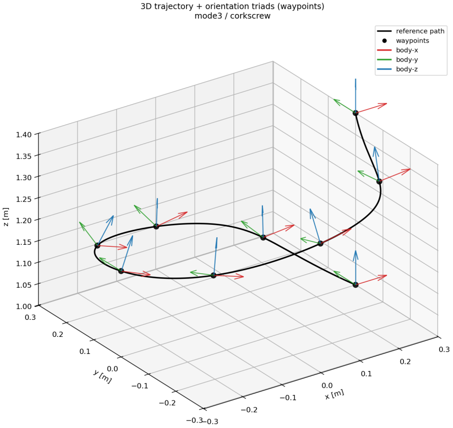
- Minimum-snap polynomial trajectories; two timing modes: Mode 1 (k_t timing), Mode 3 (time redistribution).
- Differential flatness maps the planned trajectory to the controller feedforward (thrust, attitude, $\dot{\omega}_d$).
- INDI (Tal & Karaman 2020): incremental, sensor-based control law, benchmarked on standard CF2.1.

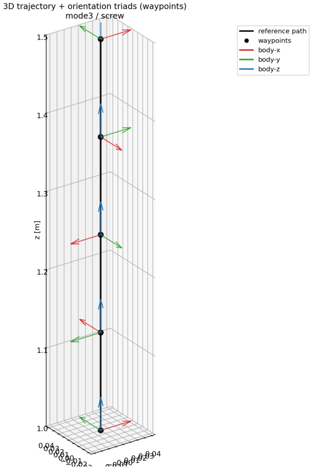
Planned Trajectories — Figure-8 / Circle / Corner



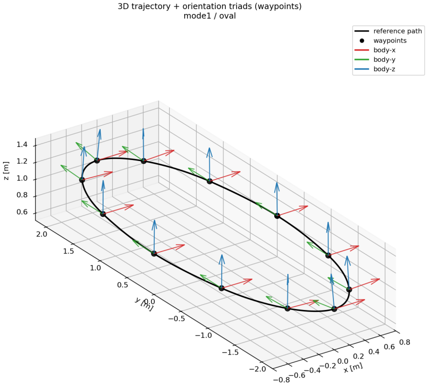


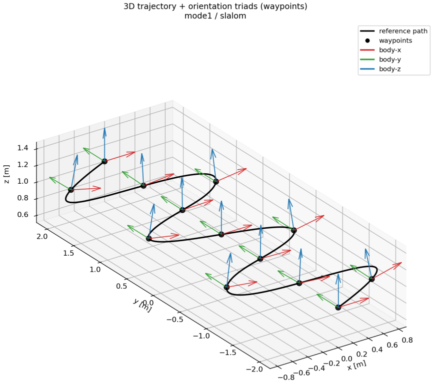
Planned Trajectories — Corkscrew / Screw



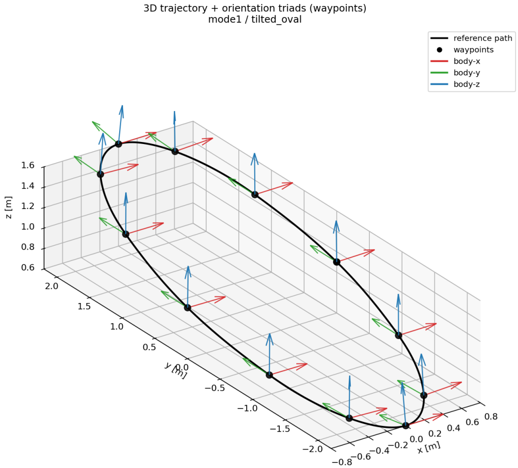


Planned Trajectories — Racing: Oval / Slalom

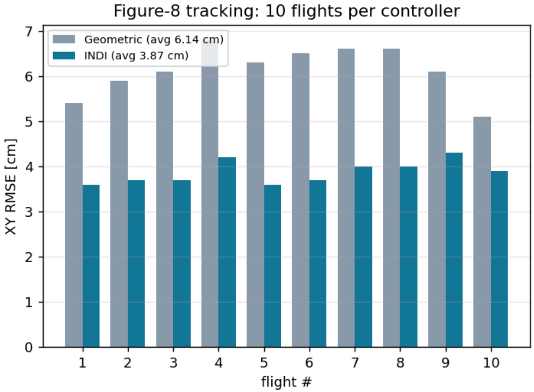


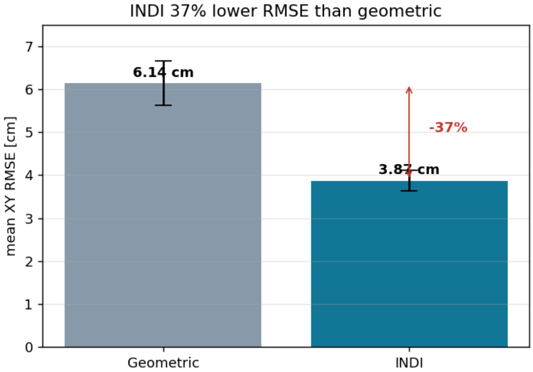


Planned Trajectories — Tilted Oval



Geometric vs. INDI — Standard CF2.1, 10 Flights Each





37% lower RMSE with INDI, consistent with the Tal & Karaman reference paper.

Figure-8 — Four-Way Comparison, Mean-Closest Flight

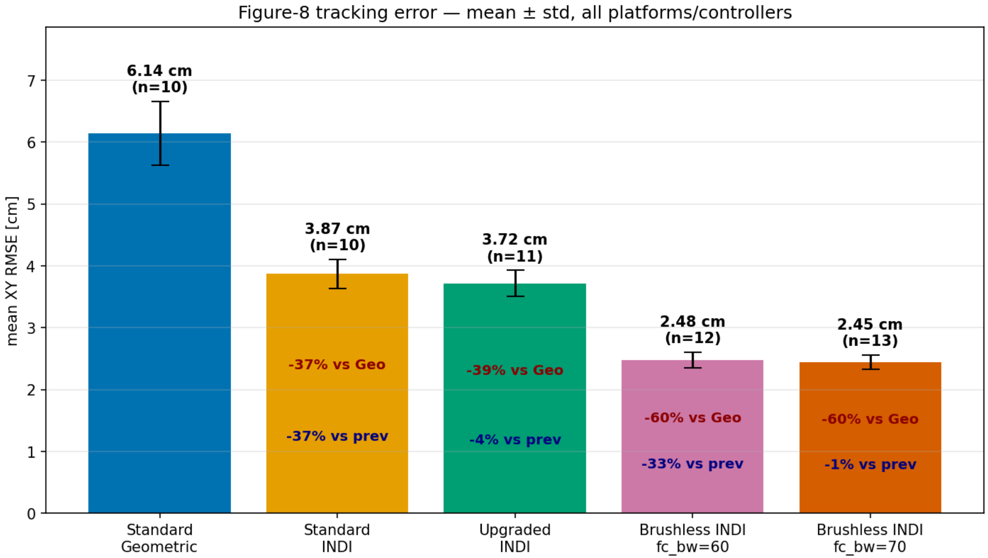
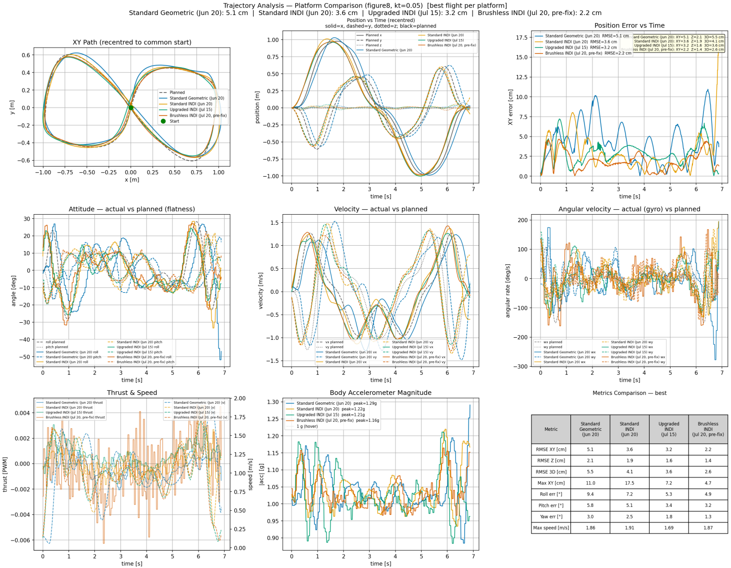


Figure-8 — Best Flight, All Four Modes: Overview



Metric	Standard Geometric (Jun 20)	Standard INDI (Jun 20)	Upgraded INDI (Jul 15)	Brushless INDI (Jul 20, pre-fix)
RMSE XY [cm]	5.1	3.6	3.2	2.2
RMSE Z [cm]	2.1	1.9	1.6	1.4
RMSE 3D [cm]	5.5	4.1	3.6	2.6
Max XY [cm]	11.0	17.5	7.2	4.7
Roll err [°]	9.4	7.2	5.3	4.9
Pitch err [°]	5.8	5.1	3.4	3.2
Yaw err [°]	3.0	2.5	1.8	1.3
Max speed [m/s]	1.86	1.91	1.69	1.87

Figure-8 — Best Flight, All Four Modes: Per-Axis

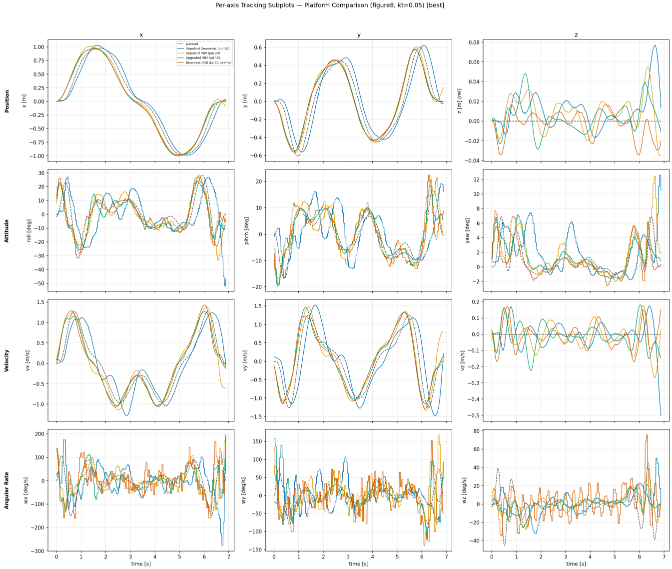


Figure-8 — Best Flight, All Four Modes: Kinematics

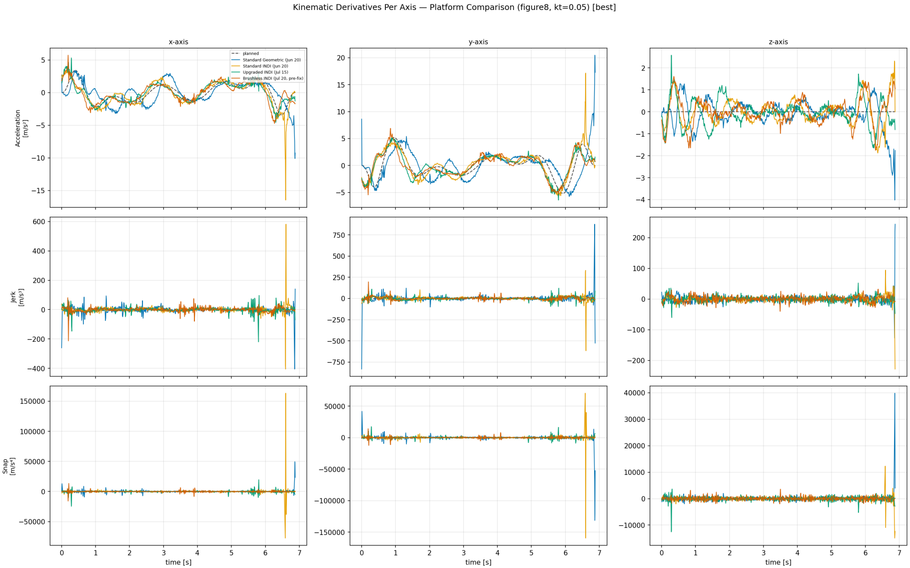


Figure-8 — Best Flight, All Four Modes: 3D Orientation

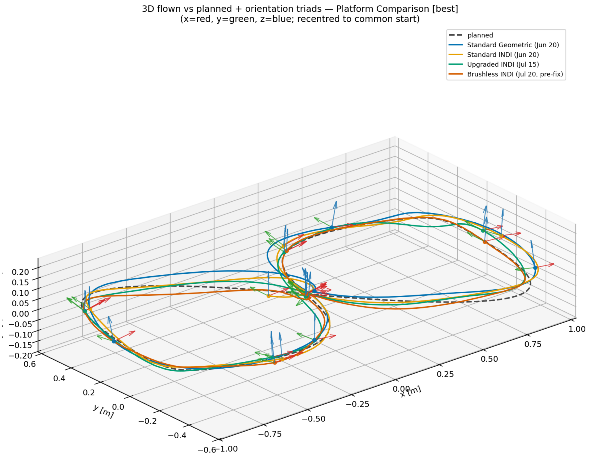
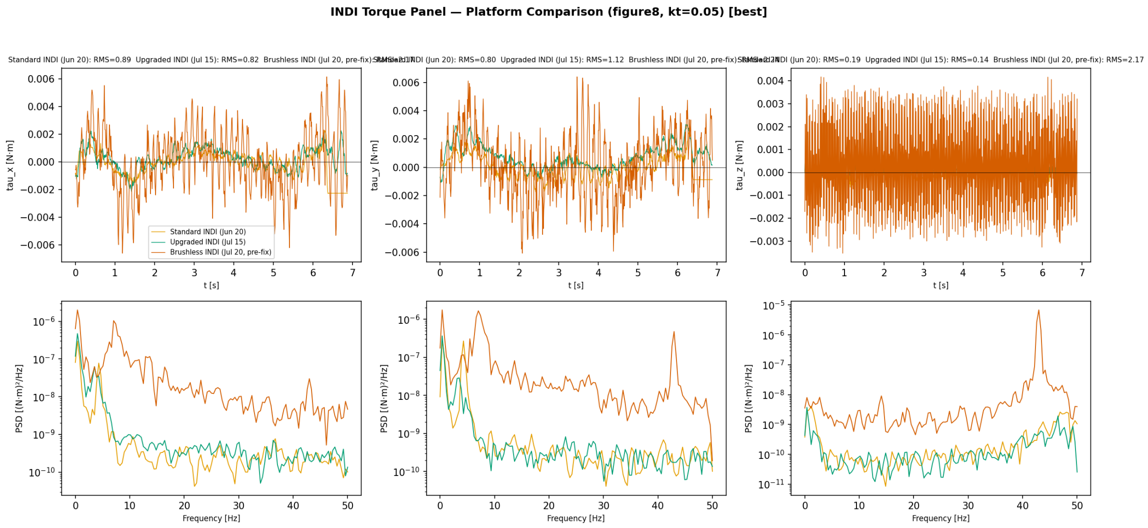


Figure-8 — Best Flight, All Four Modes: INDI Panel



No INDI telemetry (skipped): Standard Geometric (Jun 20)

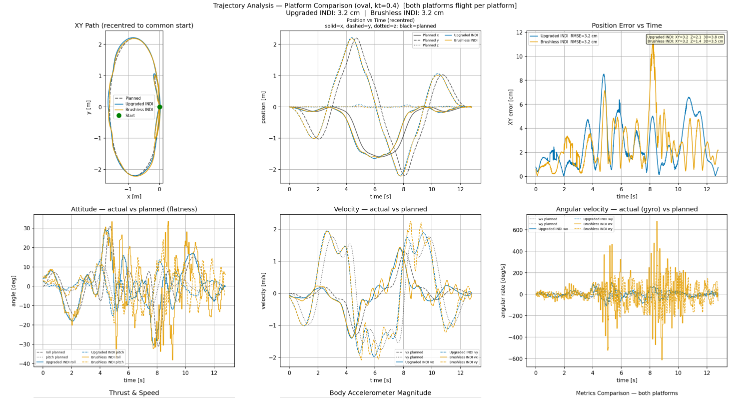
Figure-8 — Best Flight, All Four Modes: RPM Balance

Oval kt Sweep — Both Platforms

	XY / Z RMSE		Roll / Pitch / Yaw err		Max speed [m/s]	
kt	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	2.1 / 0.8 cm	2.3 / 0.5 cm	1.7/1.3/0.8°	2.2/2.9/0.9°	1.10	1.22
0.1	2.8 / 0.9 cm	1.3 / 0.7 cm	2.3/1.5/0.7°	3.1/1.6/0.3°	1.33	1.32
0.2	2.9 / 1.3 cm	2.5 / 1.1 cm	3.3/1.5/1.0°	2.6/1.3/0.4°	1.61	1.51
0.3	3.0 / 1.8 cm	2.8 / 1.0 cm	3.2/1.7/1.4°	2.9/1.5/0.5°	1.76	1.72
0.4	3.2 / 2.1 cm	3.2 / 1.4 cm	4.4/2.1/1.9°	10.5/6.1/2.0°	1.97	2.25
0.5	crash (5/5)	3.1 / 1.6 cm	—	5.5/2.7/1.4°	—	2.20
0.6	crash	not flown	—	—	—	—
0.7	not flown	crash	—	—	—	—

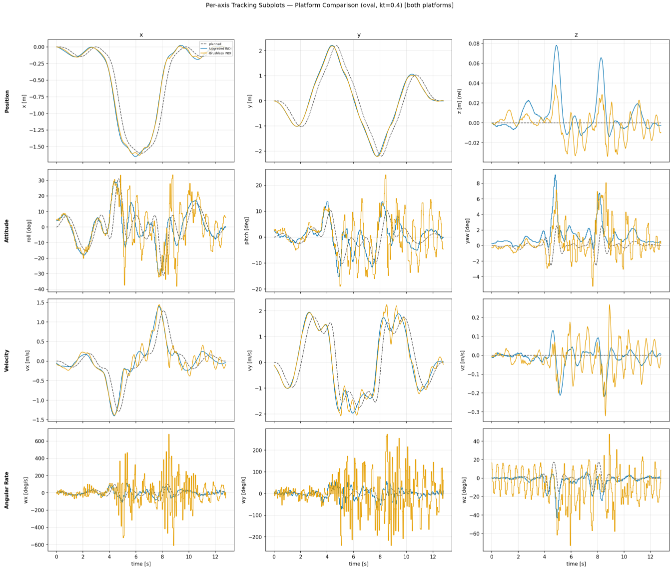
Working ceiling: upgraded ~kt 0.3–0.4, brushless ~kt 0.5–0.7.

Oval kt=0.4 — Both Platforms: Overview

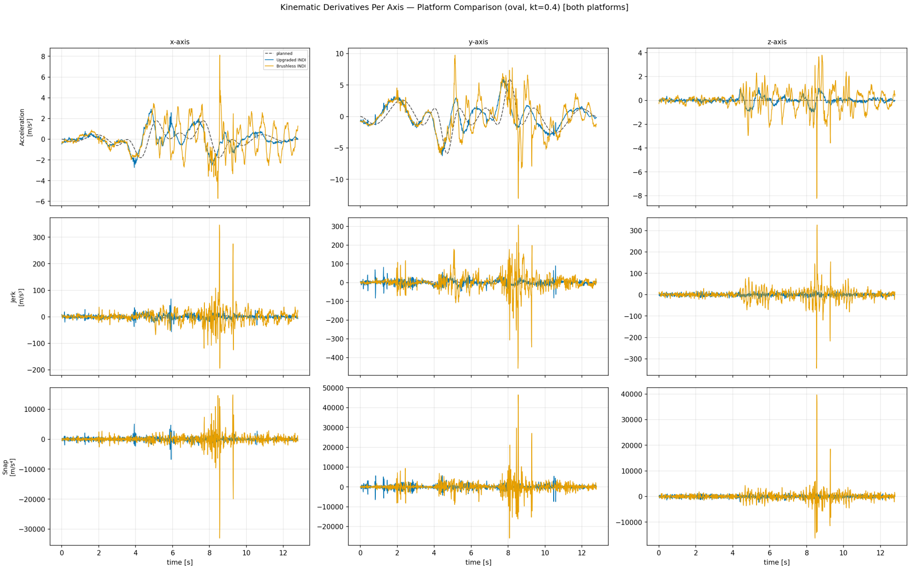


Metric	Upgraded INDI	Brushless INDI
RMSE XY [cm]	3.2	3.2
RMSE Z [cm]	2.1	1.4
RMSE 3D [cm]	3.8	3.5
Max XY [cm]	8.5	11.6
Roll err [°]	4.4	10.5
Pitch err [°]	2.1	6.1
Yaw err [°]	1.9	2.0
Max speed [m/s]	1.97	2.25

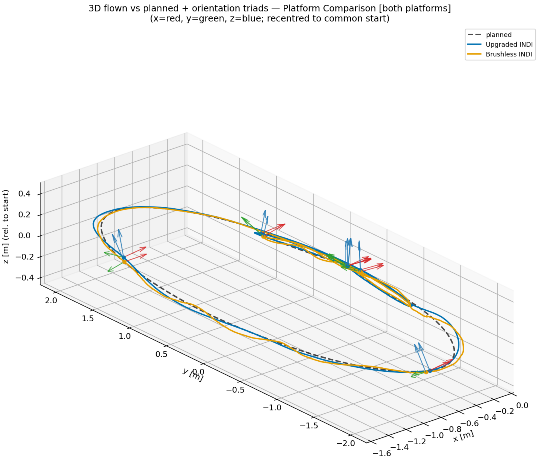
Oval kt=0.4 — Both Platforms: Per-Axis



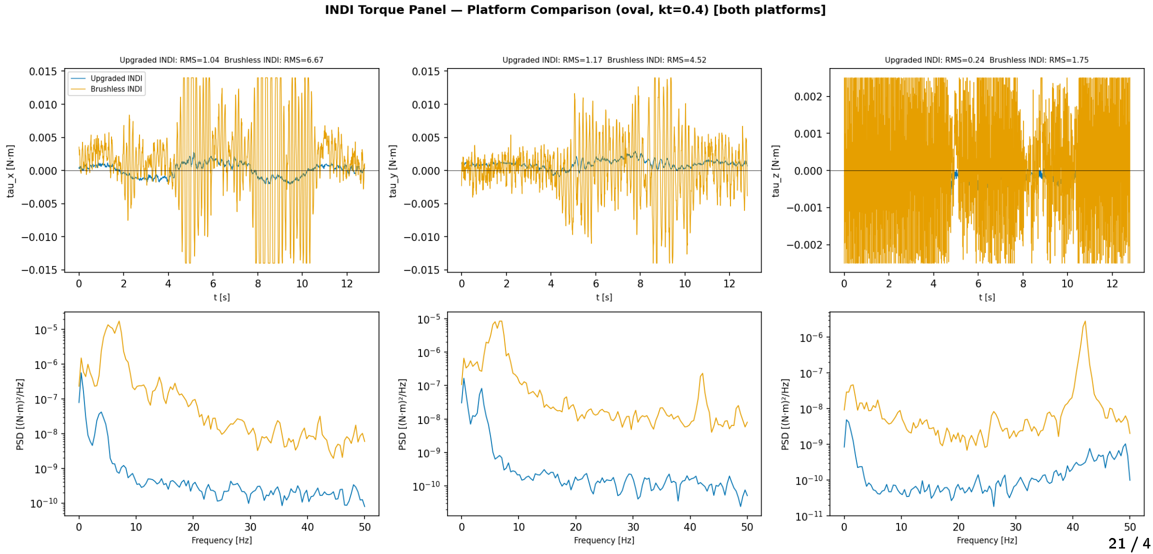
Oval kt=0.4 — Both Platforms: Kinematics



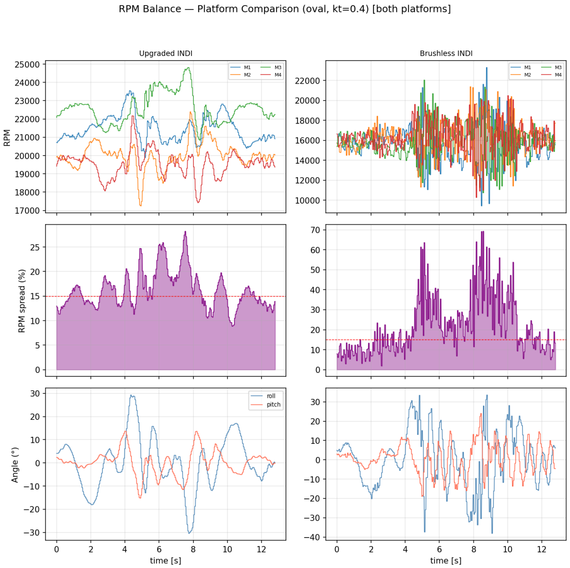
Oval kt=0.4 — Both Platforms: 3D Orientation



Oval kt=0.4 — Both Platforms: INDI Panel



Oval kt=0.4 — Both Platforms: RPM Balance

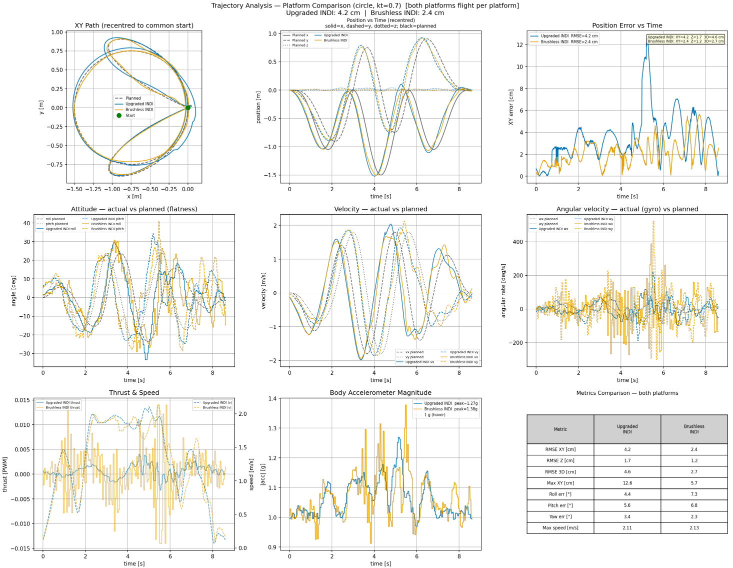


Circle kt Sweep — Both Platforms

	XY / Z RMSE		Roll / Pitch / Yaw err		Max speed [m/s]	
kt	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	1.7 / 0.8 cm	1.9 / 0.6 cm	2.1/1.8/0.8°	1.9/1.4/0.6°	1.03	1.21
0.1	1.9 / 1.3 cm	1.7 / 0.6 cm	1.9/1.8/1.1°	4.2/3.9/0.7°	1.15	1.50
0.2	2.3 / 1.2 cm	6.1 / 1.8 cm	2.2/1.8/1.4°	14.5/13.2/3.6°	1.35	1.71
0.3	2.2 / 1.5 cm	1.4 / 0.8 cm	2.3/2.1/1.9°	3.1/2.7/0.6°	1.53	1.75
0.4	2.7 / 1.8 cm	4.2 / 1.4 cm	3.5/3.6/2.5°	10.0/9.3/2.4°	1.73	1.84
0.5	2.8 / 1.6 cm	3.7 / 1.2 cm	3.5/3.5/2.6°	7.9/7.5/2.0°	1.84	1.88
0.6	3.7 / 2.4 cm	1.9 / 1.0 cm	3.1/3.3/2.7°	4.7/4.9/1.5°	1.98	1.84
0.7	4.2 / 1.7 cm	2.4 / 1.2 cm	4.4/5.6/3.4°	7.3/6.8/2.3°	2.11	2.13
0.8	3.3 / 2.5 cm	crash	4.8/5.3/3.6°	—	2.37	—
0.9	crash	crash	—	—	—	—
1.0	3.0 / 3.6 cm	not flown	4.9/4.0/4.6°	—	2.48	—

Both platforms show the same pattern: a clean ceiling, then a marginal/inconsistent zone — not a sharp wall, not a platform-specific quirk.

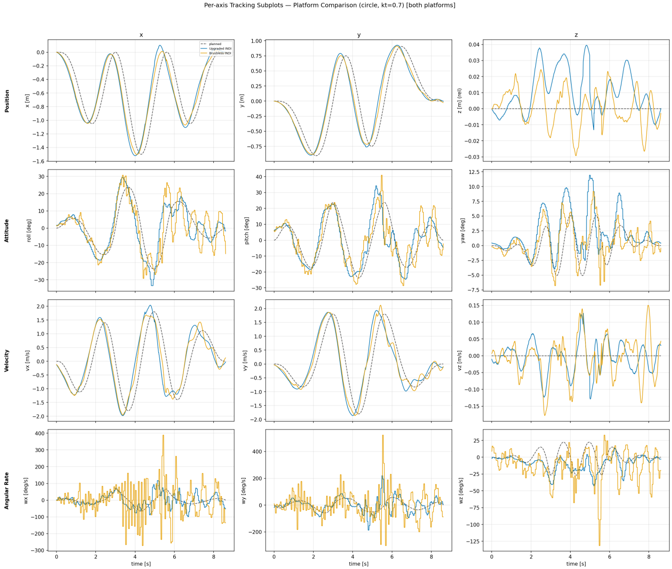
Circle kt=0.7 — Both Platforms: Overview



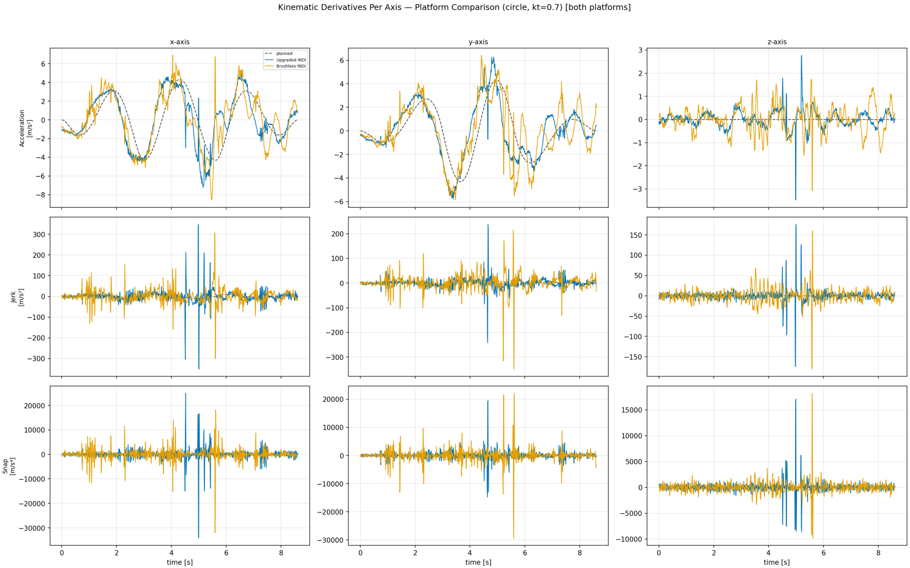
Metrics Comparison — both platforms

Metric	Upgraded INDI	Brushless INDI
RMSE XY [cm]	4.2	2.4
RMSE Z [cm]	1.7	1.2
RMSE 3D [cm]	4.6	2.7
Max XY [cm]	12.6	5.7
Roll err [°]	4.4	7.3
Pitch err [°]	5.6	6.8
Yaw err [°]	3.4	2.3
Max speed [m/s]	2.11	2.13

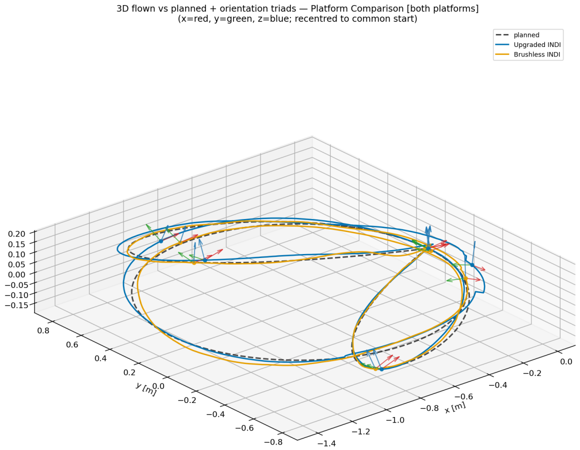
Circle kt=0.7 — Both Platforms: Per-Axis



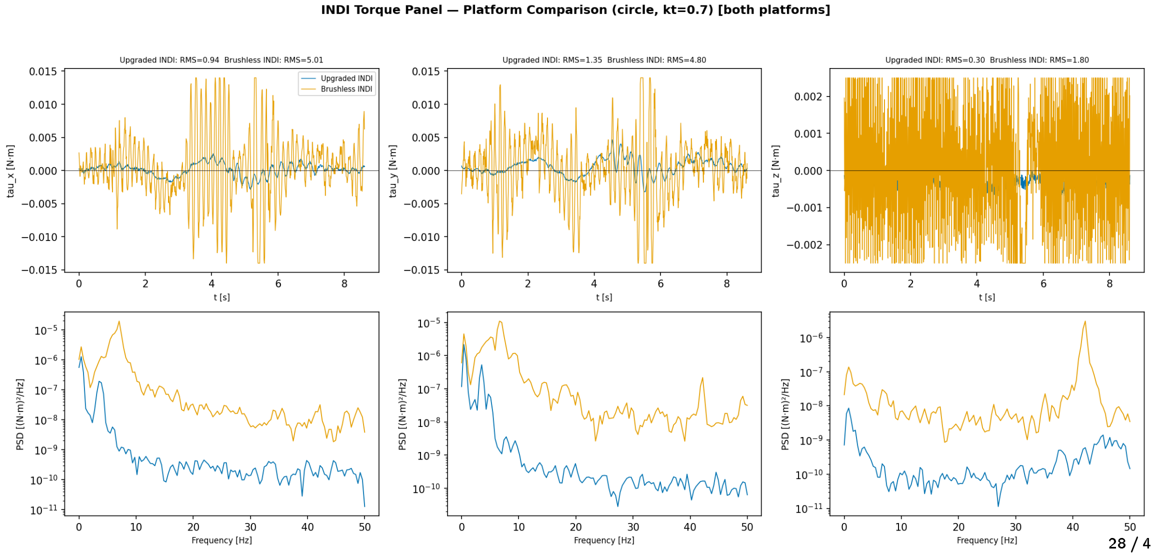
Circle kt=0.7 — Both Platforms: Kinematics



Circle kt=0.7 — Both Platforms: 3D Orientation



Circle kt=0.7 — Both Platforms: INDI Panel



Circle kt=0.7 — Both Platforms: RPM Balance

Slalom — A Different Failure Mode

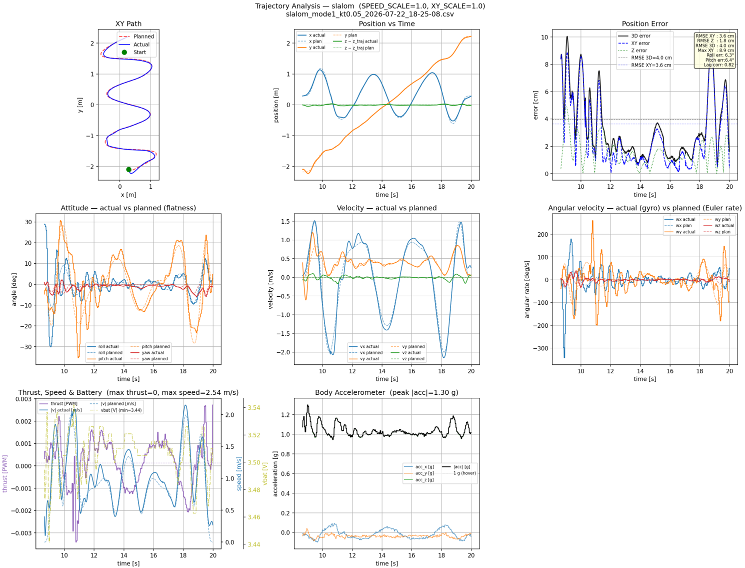
- Upgraded crashes on `slalom` at kt=0.05 — the *gentlest* speed tested — where brushless flies cleanly. Confirmed real and repeatable: **3 of 4 attempts** showed crash-level altitude drops.

- The two clearest crashes cut short early (6.5 s, 9.9 s into a longer planned flight) — the failure happens *at the first sharp direction change*, not a growing oscillation.

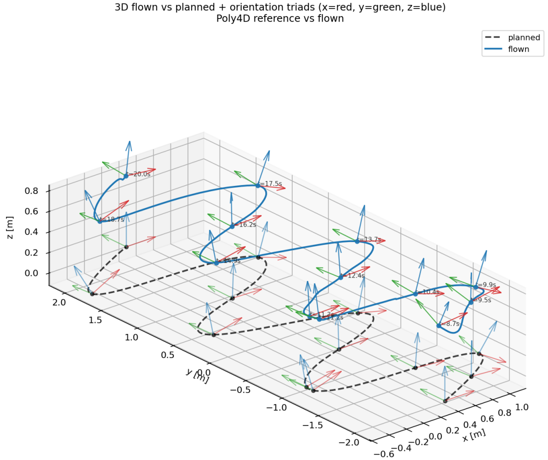
- **Not the shake mechanism**: upgraded (kr=603) never shows the sustained resonant limit cycle seen on brushless — it either tracks cleanly or crashes outright. Brushless itself still shakes at lower kr too. Root cause of the slalom-specific failure is genuinely open, not yet explained by anything else in this talk.

- No matched-flight comparison plot (upgraded has zero clean flights) — next slide shows brushless's own clean record instead.

Slalom — Brushless Flies It Cleanly



Slalom — 3D Trajectory, Flown vs Planned

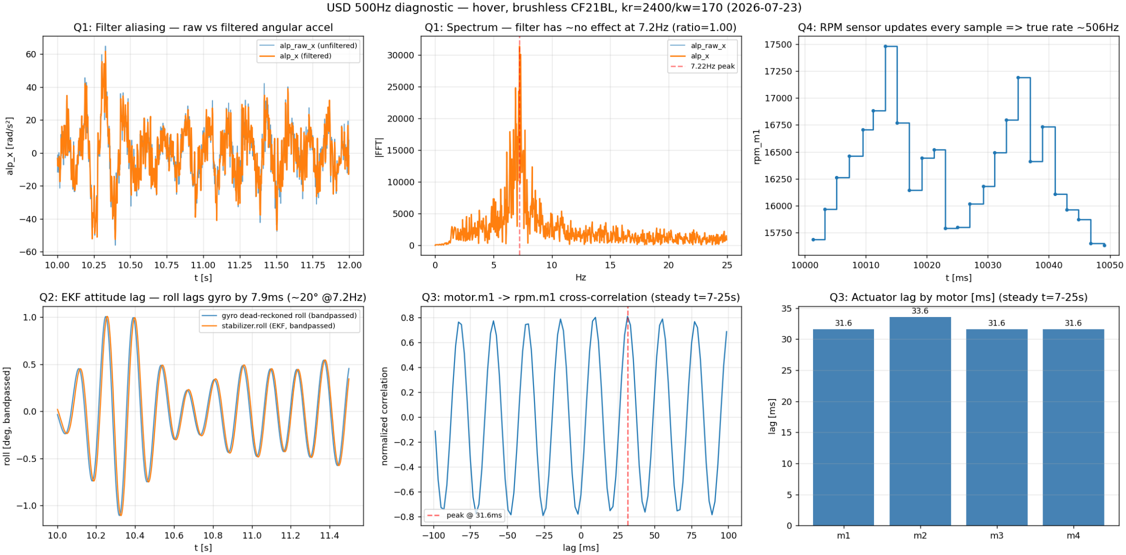


Limitations — The Brushless "Shaky Shaky"

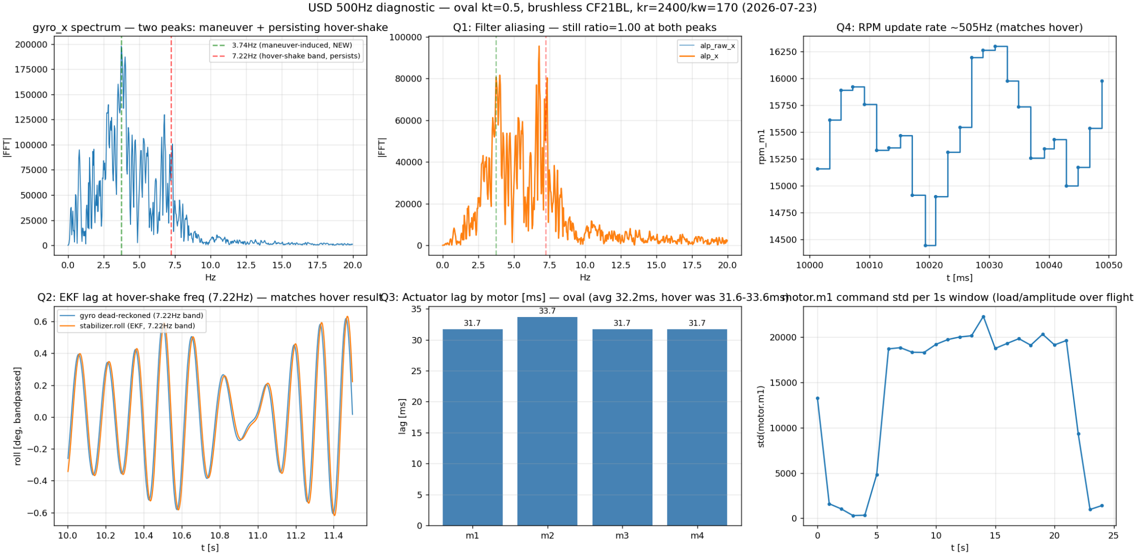
What we saw

- Full INDI on brushless CF21BL: a self-sustained ~7 Hz oscillation.
- Reproducible in hover *and* under maneuvering load — same mechanism throughout.
- **This is why brushless underperforms upgraded on some maneuvers** (e.g. crash ceiling on oval/circle at high kt) despite otherwise excellent figure-8 tracking — an unexpected, counterintuitive combination worth flagging explicitly.

Shake — Root Cause, Quantified (Hover)



Shake — Confirmed Under Real Maneuvering Load



Root Cause, Quantified — And Why the Obvious Fixes Don't Work

Factor	Measured	Verdict
Resonance $\omega_n = \sqrt{k_r}/2\pi$	7.8 Hz (kr=2400) vs. 7.22 Hz shake	**root cause**
Actuator lag @ 7.22 Hz	32 ms → **82°** phase loss	**dominant**
EKF attitude lag @ 7.22 Hz	7.9 ms → 20° phase loss	minor
Filter (fc_bw=70 Hz)	alp_filt/alp_raw ratio = 1.00	dead end
Detuning kr (≈600)	still shaky; vibration is broadband	dead end

- **Resonance**: k_r sets the attitude loop's natural frequency ($\omega_n = \sqrt{k_r}/2\pi$). At $k_r = 2400$ that's 7.8 Hz — almost exactly the 7.22 Hz shake, i.e. the controller is oscillating near its *own* natural frequency, not reacting to external vibration.
- **Actuator lag**: motor/ESC/prop response takes 32 ms to catch up to a commanded change; at 7.22 Hz that's 82° of phase shift — enough to turn a stabilizing correction into a destabilizing one one cycle later. Confirmed in-flight under real 4-motor load, not bench-only — why it's the dominant term.
- **EKF lag**: the attitude *estimate* lags true motion by only 7.9 ms (20° phase) — an order of magnitude smaller than actuator lag, ruled out as the main driver.
- **Filter**: exists specifically to remove noise like this, but at fc_bw=70 Hz it passes 7.22 Hz completely unchanged. Cutting the cutoff low enough to actually attenuate 7.22 Hz means filtering right at that frequency — reintroducing the same actuator-lag phase-loss mechanism above, right where correction authority is needed most. Same failure mode in a different place, not a fix.
- **Detuning k_r**: moves ω_n away from 7.22 Hz, but still shaky at $k_r \approx 600$ — the underlying vibration is broadband, not one clean tone, so there's no quiet k_r to retune into without giving up tracking bandwidth.

Real Fix Candidates — Correctly Not Pursued

- **Primary candidate — most consistent with the data**: a **narrow band-reject (notch) filter around 5–10 Hz**, instead of a broad low-pass. The shake is a genuine, narrow spectral peak at 7.22 Hz (hover *and* maneuvering load alike) — not broadband noise. A plain low-pass cut low enough to touch it reintroduces the same phase-lag mechanism as the actuator lag itself (Root Cause slide); a notch targeting just that band removes the resonance while leaving the rest of the spectrum — and its phase — untouched. The actual point of filtering, done narrowly instead of broadly.
- Reduce actuator lag directly: faster ESC/motor/prop response, or a delay-compensating term in the INDI control law.
- Reduce vibration at the source: prop balancing, mount damping.
- **Speculative, unproven**: motor-protocol-level rate/current limiting (e.g. DShot slew-rate limiting) to blunt the torque swings feeding the resonance — not yet backed by data collected this project.

All require hardware/firmware iteration beyond this project's remaining time — a scope decision, not a gap: diagnosis complete and quantified, fix is future work.

Limitations — Teardrop / Inverted Loop

Status: open, placeholder — attempt planned for tomorrow

- `teardrop_wide`: crashed at kt=0.05 on *both* platforms, even under plain geometric control (not just INDI) — unexplained, never root-caused.
- Inverted-loop / accel-pinned aerobatic maneuvers (loop-train, roller-coaster — Phase 4 of the original project plan): never attempted at all, blocked by the unresolved brushless shake (these maneuvers need brushless's higher thrust/omega; the upgraded drone is insufficient).
- **Plan: retry both tomorrow if time allows.** This slide will be updated with results — currently an honest placeholder, not a finished finding.

Summary

1. **Motion planning**: built a pipeline that generates optimal trajectories for the drone — the foundation everything else in this project flies on.
2. **Baseline**: established and tuned a geometric controller on the standard CF2.1 for optimal tracking error, as the reference every other mode is measured against.
3. **INDI**: implemented and deployed full INDI on **three platforms** — standard, thrust-upgraded, and brushless CF21BL.
4. **Comparison**: fair, apples-to-apples evaluation across all four modes — INDI beats geometric on every platform, and brushless beats standard/upgraded on raw tracking (best figure-8 RMSE of all four, 2.45 cm).
5. **Artifacts**: flights across the full trajectory library (figure-8, oval, circle, slalom, ...) on multiple platforms, at multiple speeds — the concrete outcome of the project.
6. **Limitations**: brushless "shaky shaky" behavior (diagnosed, not fixed) and the teardrop/inverted-loop maneuver (not attempted) — both correctly scoped out given remaining project time.

Appendix — Motion Planning, In Brief

1. **Waypoints in, smooth path out**: a handful of position (+ yaw) waypoints go in; a piecewise polynomial trajectory that passes through all of them comes out.

2. **Minimum-snap polynomials**: each segment between waypoints is a degree-7/8 polynomial, solved so the whole path is as smooth as possible (continuous up to snap, the 4th derivative of position) — this is what keeps the drone from having to make any sudden, jerky corrections.

3. **Segment timing, two ways**: either give each segment a duration by hand (Mode 0, Mellinger 2012), or give one aggressiveness scalar k_t and let the planner pick timings automatically (Mode 1, Richter et al. 2016) — this is the k_t used throughout every sweep in this talk.

4. **Differential flatness turns a path into a full reference**: position and its derivatives (velocity, acceleration, jerk, snap) plus yaw are enough to solve, in closed form, for everything the controller needs — desired attitude R_d, thrust, body rate ω_d, and angular acceleration $\dot{\omega}_d$ (Faessler, Franchi & Scaramuzza 2018). No attitude has to be authored by hand.

5. **Result**: a complete, physically consistent feedforward reference at every instant in time — not just a position setpoint — exported as polynomial coefficients and evaluated onboard at 500–1000 Hz. Both the geometric and INDI controllers consume the exact same reference.

Appendix — Full INDI, Pseudocode

```
every control cycle (dt):
- position loop (outer, full rate) -
e_p = p_d - p;  e_v = v_d - v
a_model = (∑_i k_{t,i} rpm_i^2)/mass · R e_z − g e_z   (predicted accel, from RPM² thrust model)
a_meas  = R·accel_body −g e_z   (measured accel, from IMU)
a_indi = a_meas - a_model  ← INDI increment: what the model misses
f_d = a_d + Kp e_p + Kv e_v + Ki∫e_p + g e_z + a_indi
R_d, ω_d, ω̇_des = flatness(f_d·mass, yaw_d)  (desired attitude, from planning)
- attitude loop (inner), the "full INDI" part -
e_R = 0.5 vee(R_d^T R − R^T R_d)  (attitude error on SO(3))
e_ω  = ω − ω_d
α_ref = ω̇_des − K_r e_R −K_w e_ω   (desired angular accel, Tal & Karaman Eq. 28)
α_meas = lowpass(diff(ω))  (measured angular accel, from gyro)
τ_current = rpm_to_torque(m1..m4)  (current torque, from motor RPM -- not a model)
Δτ = J (α_ref - α_meas)  ← INDI increment: correct only the error
τ = τ_current + Δτ
send(thrust, τ) → motor mixer
```

The core idea: instead of computing torque from a dynamics model alone, INDI takes the *currently measured* torque/acceleration (from RPM and gyro) as the baseline and computes only a small correction for the gap between desired and measured — far less sensitive to model error (motor dynamics, drag, mass) than a purely model-based controller.